5.7 Real Analysis

5.7.1 Closed Intervals

Proposition 5.7.1 (k-cells are closed)
Subsets of the form

$$[a_1, b_1] \times \cdots \times [a_k, b_k] \subset \mathbb{R}^k$$

are closed.

Proof. By definition of the product topology we may reduce to the case $k = 1$. Then $[a, b]$ is sequentially closed by (5.1.21), which is equivalent to being closed by (4.1.19) as $\mathbb{R}$ is first countable. ∎

5.7.2 Power Function

Proposition 5.7.2 (Power function)
Let $\alpha \in \mathbb{Q}$ and $x \in \mathbb{R}$ be non-negative numbers. Suppose $\alpha = \frac{m}{n}$ for integers m, n, where $n > 0$. Then the function

$$x^\alpha := \sqrt[n]{x^m} = \left(\sqrt[n]{x}\right)^m$$

is continuous and strictly increasing/decreasing (according to the sign of α). Furthermore

$$x^\alpha x^\beta = x^{\alpha+\beta}$$

Proof. We first consider the case $\alpha > 0$ and $m, n > 0$. The two definitions are equivalent because x^α is the unique function satisfying $(x^\alpha)^n = x^m$, the latter being an injective function. We may consider the cases $\alpha = m$ and $\alpha = \frac{1}{n}$ separately (since the composition of continuous and increasing functions is continuous and increasing). In light of (5.1.32) and (5.4.10) then it remains to show that $\sqrt[n]{x}$ is continuous. However by (5.1.31)

$$\left|\sqrt[n]{y} - \sqrt[n]{x}\right| \leq \frac{|y - x|}{n\sqrt[n]{x}^{n-1}}\,.$$

Suppose $\alpha = \frac{m}{n}$ and $\beta = \frac{p}{q}$ then $\alpha + \beta = \frac{mq+np}{nq}$ and

$$\left(x^\alpha x^\beta\right)^{nq} = (x^\alpha)^{nq}\left(x^\beta\right)^{nq} = x^{mq}x^{np} = x^{mq+np}$$

whence by uniqueness $x^\alpha x^\beta = x^{\alpha+\beta}$.

We may define $x^0 = 1$ and $x^{-\alpha} = (x^\alpha)^{-1}$, and verify that the addition formula still holds. ∎

5.7.3 Countability

Proposition 5.7.3 ($\mathbb{R}$ is separable and second countable)
The subset $\mathbb{Q} \subset \mathbb{R}$ is dense. Furthermore $\mathbb{R}$ is second countable, with a base formed by open balls with rational center and radius.

Similarly $\mathbb{Q}^k$ is dense in $\mathbb{R}^k$, which is second countable.

Proof. We only need to show that $\mathbb{Q} \subset \mathbb{R}$ is dense, and the rest follows from (5.3.15) because $\mathbb{Q}$ is countable.

By (5.1.26) for every $x \in \mathbb{R}$ there is a sequence $x_n \in \mathbb{Q}$ such that $x_n \to x$. Then (4.1.19) shows that $x \in \overline{\mathbb{Q}}$ as required. ∎

5.7.4 Bolzano-Weierstrass Theorem

Proposition 5.7.4 (Monotone Subsequence Theorem)
Let (a_n) be a sequence in an ordered field. Then it has a monotone subsequence.

Proof. Define the set of peaks to be

$$X = \{n \in \mathbb{N} \mid m \geq n \implies x_n \geq x_m\}$$

If X is infinite then the elements determine a decreasing subsequence. Suppose X is finite and define $n_1 := \max(X) + 1$. We claim inductively that there is a sequence of integers

$$n_1 < n_2 < \cdots < n_k < \cdots$$

such that a_{n_k} is increasing. Evidently $n_k \notin X$ whence there exists $n_{k+1} > n_k$ such that $a_{n_{k+1}} > a_{n_k}$. ∎

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Proposition 5.7.5 (Bolzano-Weierstrass Theorem)
Let (a_n) be a bounded sequence in $\mathbb{R}$. Then it contains a convergent subsequence.

Proof 1.. By (5.7.4) a_n has a monotone subsequence. By (5.1.23) this subsequence is convergent. ∎

Proof 2.. Define two sequences of real numbers (l_n), (u_n) recursively as follows

$$(l_n, u_n) := \begin{cases} (L, U) & n = 1 \\ \left(l_{n-1}, \frac{l_{n-1}+u_{n-1}}{2}\right) & \{a_n\} \cap \left[l_{n-1}, \frac{l_{n-1}+u_{n-1}}{2}\right] \text{ is infinite} \\ \left(\frac{l_{n-1}+u_{n-1}}{2}, u_{n-1}\right) & \text{otherwise} \end{cases}$$

Then by definition

a) (l_n) is increasing and (u_n) is decreasing
b) $l_n \leq u_m$ for all n, m
c) $|u_n - l_n| = \frac{U-L}{2^n} \to 0$
d) $[l_n, u_n]$ contains infinitely many elements of the sequence (a_n)

We claim that

$$\bigcap_{n\in\mathbb{N}} [l_n, u_n] = \{\sup l_n\} = \{\inf u_n\}$$

Suppose x, y lie in the intersection then by c) we have $|x - y| < \varepsilon$ for all ε, whence they are equal. So the set consists of at most one element. We claim $\sup l_n$ lies in the intersection. For $l_n \uparrow \sup l_n$ by (...) and $l_n \leq u_m$ whence $\sup l_n \leq u_m$ by (...) as required.

Finally let $\left(a_{n_k}\right)$ be a subsequence such that $a_{n_k} \in [l_k, u_k]$ consequently by c) we have $\left|a_{n_k} - \alpha\right| \to 0$ where $\alpha := \sup l_n$. ∎

Corollary 5.7.6 (Bolzano-Weierstrass in $\mathbb{R}^k$)
Let (a_n) be a bounded sequence in $\mathbb{R}^k$. Then it contains a convergent subsequence.

Proof. Observe that $(a_n) = \left(a_n^1, \cdots, a_n^k\right)$ where a^i is a bounded sequence in $\mathbb{R}$. By applying (5.7.5) to each coordinate in turn we may find a subsequence n_j such that $a^i_{n_j} \to a^i$ for $i = 1 \cdots k$. By definition of the product norm it is clear that $a_{n_j} \to \left(a^1, \cdots, a^k\right)$. ∎

Proposition 5.7.7 (Heine-Borel Theorem)
Let X be a subset of $\mathbb{R}^k$. Then the following are equivalent

a) X is closed and bounded
b) X is sequentially compact
c) X is compact

In particular every closed k-cell $[\boldsymbol{a}, \boldsymbol{b}]$ is compact.

Proof 1.. Recall from (5.7.3) that $\mathbb{R}^k$ is second countable, and therefore so is X. Therefore the equivalences $b) \iff c)$ follows from (4.1.106).

$b), c) \implies a)$ We may use (4.1.110) to show that X is closed, observing that $\mathbb{R}^k$ is both first countable and Kolmogorov. Alternatively we may observe $\mathbb{R}^k$ is Hausdorff and appeal to (4.1.107).

To show that it is bounded we may assume it is not to find a sequence x_n such that $\|x_n\| \geq n$. By assumption this has a convergent subsequence $x_{n_j} \to x$. Further there is some N_1 such that $N_1 > \|x\|$ and some N_2 such that $n \geq N_2 \implies \|x_n - x\| < N_1 - |x| \implies \|x_n\| < N_1$. Then taking $n = \max(N_1, N_2)$ implies $\|f(x_n)\|$ is both less and greater than N_1, a contradiction.

Alternatively we may reduce to the case $k = 1$ by noting that $\pi_i(X)$ is compact (4.1.109). The family $\{B(n; 1) \mid n \in \mathbb{N}\}$ forms an open cover, for which there must exist a finite subcover. This immediately shows that $\pi_i(X)$, and therefore X, is bounded.

$a) \implies b), c)$ By assumption $X \subset [-M, M] \times \cdots \times [-M, M]$ for some M, which by (5.7.6) is sequentially compact. We may then use either (4.1.108) or (4.1.111). ∎

Proof 2.. Firstly X is complete iff it is closed by (5.3.13). Evidently a subset of $\mathbb{R}$ is totally bounded iff it is bounded, and the same argument may be extended to $\mathbb{R}^k$. Therefore the equivalence follows from (5.3.17). ∎

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5.7.5 Boundedness Theorem

Proposition 5.7.8
Let X be a compact topological space and $f : X \to \mathbb{R}^k$ be a continuous function. Then f is bounded and attains its bounds.

Proof 1.. We may prove directly the case $X = [a, b]$ and $k = 1$.

Suppose f is not bounded, then there is some sequence (x_n) in X such that $|f(x_n)| \geq n$. By choosing a convergent subsequence (5.7.5) we may assume that $x_n \to x$. Further X is closed so $x \in X$ (4.1.19). By continuity (5.4.13) $f(x_n) \to f(x)$. Choose $N_1 > |f(x)|$ and N_2 such that $n \geq N_2 \implies |f(x_n) - f(x)| < N_1 - |f(x)| \implies |f(x_n)| < N_1$. Then $n := \max(N_1, N_2) \implies |f(x_n)| < N_1$ and $|f(x_n)| > N_1$ a contradiction. ∎

Proof 2.. We may reduce to the case $k = 1$ by considering the composition with the norm $\|\cdot\|$.

By (4.1.109) (or (4.1.112)) $f(X)$ is (sequentially) compact. Then by (5.7.7) $f(X)$ is closed and bounded.

Let $y = \sup f(X)$. By (5.1.18) there exists $y_n \to y$ with $y_n \in f(X)$. Since $f(X)$ is closed we see $y \in f(X)$ by (4.1.19) as required. ∎

5.7.6 Intermediate Value Theorem

Proposition 5.7.9 (Intervals are connected)
Let $X \subset \mathbb{R}$. Then the following are equivalent
a) X is connected
b) For every $x < y < z$ such that $x, z \in X$, we have $y \in X$

In particular $(a, b], (a, b), [a, b)$ are connected.

Proof. $b) \implies a)$ Suppose X is disconnected, then $X = U \sqcup V$ with U, V non-empty open and closed subsets of X. Let $x \in U$ and $z \in V$, and assume wlog that $x < z$. Define

$$\alpha := \sup(U \cap (x, z))$$

By (5.1.21) $\alpha \in [x, z] \subset X$. By (5.1.18) there exists $\alpha_n \in U \cap (x, z)$ such that $\alpha_n \uparrow \alpha$. Then by (4.1.20) we have $\alpha \in \mathrm{cl}_{\mathbb{R}}(U) \cap X \overset{(4.1.21)}{=} \mathrm{cl}_X(U) \overset{(4.1.19)}{=} U$. If $\alpha = z$ then we reach a contradiction as U and V were assumed to be disjoint. Similarly if $\alpha < z$, as U is open in X, we may choose $0 < \varepsilon < z - \alpha$ such that $(\alpha - \varepsilon, \alpha + \varepsilon) \cap X \subset U$. By assumption $\alpha + \varepsilon \in X$ whence $\alpha + \varepsilon \in U$, which contradicts maximality of α.

$a) \implies b)$ Suppose we have $x < y < z$ such that $x, z \in X$ but $y \notin X$. Then define the open subsets $U = (-\infty, y) \cap X$ and $V = (y, \infty) \cap X$. Then

$$U := (-\infty, y) \cap X = (-\infty, y] \cap X$$

is both open and closed. ∎

Proposition 5.7.10 (Intermediate Value Theorem)
Let $a < b$ be real numbers and $f : [a, b] \to \mathbb{R}$ continuous such that $f(a) < f(b)$. Then for every $f(a) < c < f(b)$ there exists $x \in (a, b)$ such that $f(x) = c$.

Proof 1.. Consider

$$X := \{x' \in [a, b] \mid f(x') < c\}$$

By definition X is bounded, and $a \in X$, so it is non-empty, so we may define $x := \sup X$. We claim that $f(x) = c$. By (5.1.18) there is a sequence $x_n \in X$ such that $x_n \to x$. By (5.4.13) $f(x_n) \to f(x)$ and by (5.1.21) $x \leq b$ and $f(x) \leq c$. Further by assumption $x \neq b$.

To show $f(x) \geq c$ consider the sequence $x_n := x + \min\left(\frac{1}{n}, b - x\right) \in [a, b]$. Evidently $x_n \to x$ so by continuity $f(x_n) \to f(x)$. By construction $f(x_n) \geq c$ whence $f(x) \geq c$.

Therefore we conclude $f(x) = c$ as required. ∎

Proof 2.. By (5.7.9) $[a, b]$ is connected, whence by (4.1.118) $f([a, b])$ is connected. The result follows from (5.7.9) again. ∎

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5.7.7 Mean-Value Theorem

Proposition 5.7.11 (Mean-Value Theorem)
Let $f : [a, b] \to \mathbb{R}$ be a continuous function which is differentiable on (a, b). Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof 1.. Define the function

$$g(x) := f(x) - \frac{f(b) - f(a)}{b - a}(x - a)$$

Then g satisfies the same conditions with $g(a) = g(b) = f(a)$ and $g'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}$. Evidently it is sufficient to find $c \in (a, b)$ such that $g'(c) = 0$.

By (5.7.8) g is bounded and attains its maximum and minimum, say at c, d respectively. If both $c, d \in \{a, b\}$ then g must be the constant function whence g' is exactly zero. Otherwise we may assume for example that $c \in (a, b)$. By (5.5.2)

$$\lim_{n\to\infty} \frac{g\left(c \pm \frac{1}{n}\right) - g(c)}{\pm\frac{1}{n}} = g'(c)$$

As c is an extremum we may deduce from (5.1.21) that both $g'(c) \leq 0$ and $g'(c) \geq 0$, whence $g'(c) = 0$. ∎

Corollary 5.7.12
Let $f : [a, b] \to \mathbb{R}$ a continuous function which is differentiable on (a, b). If $f'(x)$ is positive (resp. strictly positive, negative, strictly negative) then $f(x)$ is increasing (resp. strictly increasing, decreasing, strictly decreasing).

Proposition 5.7.13
Let $f : (a, b) \to \mathbb{R}$ be a strictly increasing (resp. decreasing) continuous function. Then f is a homeomorphism onto its image.

Proof. Evidently f is injective, therefore it is only required to show that the map

$$f^{-1} : f(U) \to U$$

is continuous. Without loss of generality we may assume that f is strictly increasing. Suppose $y = f(x)$ then we require to show for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|y' - y| < \delta \implies \left|f^{-1}(y') - f^{-1}(y)\right| < \varepsilon$$

Define $\varepsilon' \leq \varepsilon$ such that $(x - \varepsilon', x + \varepsilon') \subset (a, b)$. Define $\delta := \min(f(x + \varepsilon') - f(x), f(x) - f(x - \varepsilon'))$. Then

$$|y' - y| < \delta \implies f(x - \varepsilon') < y' < f(x + \varepsilon') \implies x - \varepsilon' < f^{-1}(y') < x + \varepsilon' \implies \left|f^{-1}(y') - f^{-1}(y)\right| < \varepsilon' \leq \varepsilon$$

as f^{-1} is also monotonic. ∎

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